Leif Adamec Rydenfalk

Umeå, Sweden

ledamecrydenfalk@gmail.com · LinkedIn · GitHub · adamec.me

SUMMARY

Software engineer who ships end-to-end — from zero-copy Rust runtimes and real-time GPU renderers to multi-platform SaaS products. Equally at home in deep systems work (a 1.48M-msg/s distributed substrate, a P2P compute economy) and in shipping polished products solo (a marketplace and an EdTech SaaS, both live in production). Self-taught; learns from first principles and builds the whole stack.

EXPERIENCE

Systems Engineer — Internal AI Tools · Gaming Studio

March 2025 – Present · Umeå, Sweden

- Built and shipped an autonomous "time-guardian" tool in Rust (Axum/Tokio) that ingests employee time reports from Fortnox, analyzes them against company policy with an LLM, and nudges employees over Slack — turning a manual, multi-day reconciliation chore into a self-running service.
- Reduced payroll reconciliation time from ~3 days to ~30 minutes; runs unattended with automatic OAuth token refresh and autonomous Fortnox↔Slack identity mapping.
- Secured an admin dashboard behind Google OAuth; wrote property-based tests for the validation engine that caught 11 edge-case bugs pre-production.
- Deployed from a locally hosted Mac Mini via Cloudflare Tunnels with zero downtime in production.

Independent Software Engineer / Founder · Rydenfalk Systems

February 2025 – Present · Umeå, Sweden

- **Myra / Trana** (trana.app) — designed and built an AI lesson-material SaaS for Swedish upper-secondary teachers, solo. One TypeScript monorepo (Turborepo, pnpm) ships four clients from shared packages: web (TanStack Start + React 19 on Cloudflare Workers), landing (Astro on Cloudflare Pages), mobile (Expo, App Store + Play Store), and desktop (Electron). Supabase backend with row-level security; full test pyramid (Vitest, Playwright, pgTAP, Deno) and per-app CI/CD.
- **Cell** — a biologically-inspired distributed computing substrate in Rust: sandboxed "cells" exchange zero-copy, rkyv-archived messages over Unix sockets. Sustains **1.48M messages/sec at ~677ns median RTT on a single core**, with an embeddable Raft consensus layer and a 9M-TPS market-simulation benchmark. Underpins **CE**, a P2P compute economy (libp2p NAT traversal, custom PoW credit ledger, gVisor sandboxing) with a public relay at ce-net.com, packaged for Homebrew, AUR, Scoop, and Chocolatey.
- **OpenJaws** — a distributed "cell mesh" platform (TypeScript + Rust) where self-discovering cells gossip-sync a shared atlas and expose 100% type-safe capabilities through a router/procedure API. Add a directory with a Cell.toml, it joins the mesh; remove it, the mesh heals — zero service-mesh configuration.
- **Blueberry Browser** — an Electron/Chromium browser that exposes a Model Context Protocol (MCP) server, letting AI agents operate a real, logged-in session like a human. Used daily to automate Gmail, WhatsApp Web, and LinkedIn with no per-site integrations. Concept to daily-use tool in two weeks.

Full-Stack Contractor · Self Employed

January 2024 – May 2025 · Umeå, Sweden

- Delivered commercial codebases (SvelteKit + Node.js + PostgreSQL) with CI/CD via GitHub Actions.
- Integrated Stripe Connect for split payments and handled PCI-sensitive flows via Stripe webhooks.
- Containerized all services with Docker; dev–prod parity under a 2-minute cold start on a new machine.
- Built a personalized feed-generation algorithm driven by user attributes.

SELECTED PROJECTS

Myra / Trana trana.app

AI lesson-material SaaS for Swedish teachers. Single TypeScript monorepo → web, landing, mobile, and desktop from shared ui/config packages. TanStack Start + React 19, Astro, Expo, Electron, Supabase (RLS, Edge Functions). Tested with Vitest, Playwright, and pgTAP; shipped through per-app CI/CD to Cloudflare, the App Store/Play Store, and GitHub Releases.

Rheo rheo.se

Mobile-first resale marketplace, ~2M lines of Rust. Axum services, React Native client, PostgreSQL, Redis Streams, Stripe Connect, PostNord shipping, Cloudflare R2. Multi-currency design with a per-listing "anchor currency" so prices never drift from what the seller chose. Three independent monitoring layers (in-process Sentinel + Cloudflare Worker + Hetzner systemd timer) that fail independently and alert over email/Telegram. p95 ~78ms on a 256MB container; blue-green deploys.

Cell / CE github.com/Leif-Rydenfalk/cell

Biologically-inspired distributed substrate in Rust — "cells," "membranes," "vesicles," "synapses." Zero-copy rkyv messaging at 1.48M msg/s / ~677ns RTT, embeddable Raft consensus, #[protein] codegen macros. Foundation for CE, a P2P compute economy (libp2p, PoW credit ledger, gVisor sandboxing).

OpenJaws + Cell Mesh Protocol github.com/Leif-Rydenfalk/openjaws

Distributed cell-mesh runtime. Sovereign cells self-discover via gossip-based atlas sync, self-replicate via Cell.toml, and expose typed RPC with Ed25519 capability proofs. Reference SDK open-sourced as cell-mesh-protocol-1.

Blueberry Browser github.com/Leif-Rydenfalk/blueberry-browser
Electron browser with a built-in MCP server. AI agents delegate real web-UI tasks (clicks, form fills, reads) to a live, logged-in Chromium session — no per-site integrations or API tokens.

Voxel Game Engine github.com/Leif-Rydenfalk/game-engine
Rust/wgpu engine as a Cargo workspace: a reusable engine library (render graph, ECS via hecs, hot-reloaded WGSL shaders, audio, input) driving a SDF-voxel ray-traversal renderer with TAA, bloom, and atmospheric scattering at 60 FPS on Apple A15/M1 — plus a separate N-body orbital ("spacetime") simulator on the same engine.

Also: Dream Engine (real-time biologically-inspired neuron/learning sim, 90 FPS on a GTX 1060), Mariana (zero-dependency streaming LLM reasoning console), CaptureFlow (production AI assistant with web + Telegram), a chess engine, and a deep back-catalog of graphics/simulation experiments in Rust and C++.

SKILLS

Languages: Rust (advanced), TypeScript, JavaScript, Python, SQL, WGSL, C, C++

Backend: Axum, Tokio, Node.js, REST/WebSockets, microservices, Redis Streams, PostgreSQL, Supabase

Frontend: SvelteKit, React/React 19, React Native (Expo), TanStack Start, Astro, Dominator, Electron, Tailwind CSS

Distributed Systems: libp2p, PoW blockchain, Raft consensus, gossip protocols, zero-copy IPC (rkyv), Ed25519 identity, NAT traversal

Graphics: wgpu, WGSL shaders, SDF/ray-marching, voxel traversal, ECS (hecs), TAA, bloom, render graphs

AI / Agents: MCP (Model Context Protocol), Claude & Gemini APIs, LLM tool-use, Chromium/Electron automation

Infrastructure: Docker, gVisor, Cloudflare (Workers, Pages, Tunnels, R2), Hetzner, Railway, GitHub Actions, blue-green deploys, package distribution (Homebrew, AUR, Scoop, Chocolatey)

Testing: TDD, property-based testing, Playwright/Vitest E2E, pgTAP, integration testing

EDUCATION

Self-taught.